



Shore

Year 12
Trial HSC Examination
August 2016

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Board-approved calculators may be used
- A BOSTES Reference Sheet is provided
- Answer Questions 1–10 on the Multiple Choice Answer Sheet provided
- In Questions 11–16, show relevant mathematical reasoning and/or calculations
- Start each of Questions 11–16 in a new writing booklet
- Write your examination number on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A” on the front cover

Examination Number:

Set:

Total marks – 100

Section I Pages 2 – 5

10 marks

- Attempt questions 1–10
- Allow about 15 minutes for this section

Section II Pages 6 – 13

90 marks

- Attempt questions 11–16
- Allow about 2 hours and 45 minutes for this section

Note: Any time you have remaining should be spent revising your answers.

Section I

10 marks

Attempt Questions 1–10

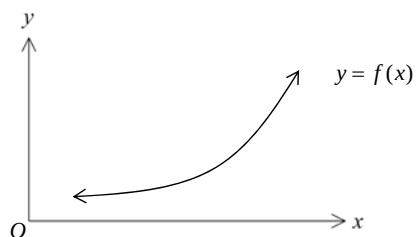
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 A computer costs \$1364.00 including 10% GST. What is the pre-GST cost of the computer?
(A) \$136.40
(B) \$1227.60
(C) \$1240.00
(D) \$1500.40
- 2 The quadratic equation $x^2 + 5x - 2 = 0$ has roots α and β . What is the value of $\alpha\beta - (\alpha + \beta)$?
(A) 3
(B) 7
(C) -3
(D) -7
- 3 What is the equation of the locus of a point that is always 5 units from the point $(2, -3)$?
(A) $(x - 2)^2 + (y + 3)^2 = 5$
(B) $(x + 2)^2 + (y - 3)^2 = 25$
(C) $(x + 2)^2 + (y - 3)^2 = 5$
(D) $(x - 2)^2 + (y + 3)^2 = 25$

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

- 4 $y = f(x)$ is shown on the number plane.



Which of the following statements is true?

- (A) $y = f(x)$ is decreasing and concave up.
 (B) $y = f(x)$ is decreasing and concave down.
 (C) $y = f(x)$ is increasing and concave up.
 (D) $y = f(x)$ is increasing and concave down.

- 5 What are the solutions to $2\sin x = -\sqrt{3}$ for $0 \leq x \leq 2\pi$?

- (A) $\frac{\pi}{3}$ and $\frac{2\pi}{3}$
 (B) $\frac{4\pi}{3}$ and $\frac{5\pi}{3}$
 (C) $\frac{\pi}{3}$ and $\frac{5\pi}{3}$
 (D) $\frac{4\pi}{3}$ and $\frac{2\pi}{3}$

- 6 If $a = \log_5 2$ and $b = \log_5 3$, what expression is equivalent to $\log_5 36$?

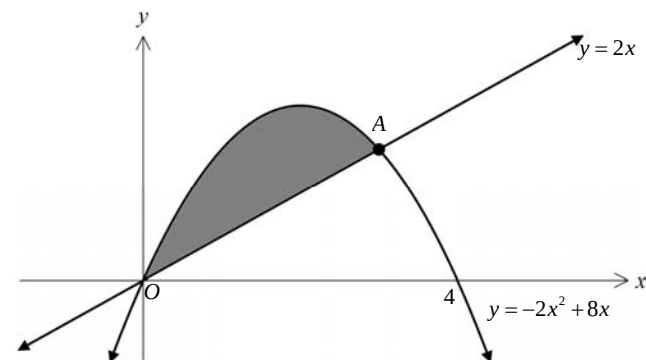
- (A) $2(a+b)$
 (B) $2ab$
 (C) $a^2 + b^2$
 (D) $(ab)^2$

- 7 The limiting sum of the geometric series $4 + 8x + 16x^2 + 32x^3 + \dots$ is 140.

What is the value of x ?

- (A) $\frac{18}{35}$
 (B) $\frac{1}{2}$
 (C) $-\frac{1}{70}$
 (D) $\frac{17}{35}$

- 8 The parabola $y = -2x^2 + 8x$ and the line $y = 2x$ intersect at the origin and at point A.



Which expression could be used to calculate the area enclosed by the parabola and the line?

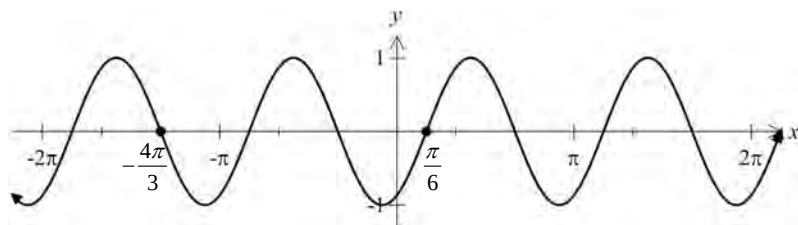
- (A) $\int_0^4 -2x^2 + 6x \, dx$
 (B) $\int_0^4 -2x^2 + 8x \, dx - \int_0^3 2x \, dx$
 (C) $\int_0^3 -2x^2 + 6x \, dx$
 (D) $\int_0^3 -2x^2 + 8x \, dx - \int_0^4 2x \, dx$

- 9 The numbers 1 to 20 are written on cards and placed in a bag. One card is drawn at random.

What is the probability that the number on the card is even or a multiple of 3?

- (A) $\frac{1}{2}$
 (B) $\frac{4}{5}$
 (C) $\frac{13}{20}$
 (D) $\frac{7}{20}$

- 10 The graph shows the equation $y = \sin(Ax - B)$ over the domain $-2\pi \leq x \leq 2\pi$.



What are the values of A and B ?

- (A) $A = 2, B = \frac{\pi}{3}$
 (B) $A = \frac{1}{2}, B = \frac{\pi}{3}$
 (C) $A = 2, B = \frac{\pi}{6}$
 (D) $A = \frac{1}{2}, B = \frac{\pi}{6}$

Section II

90 marks

Attempt Questions 11–16

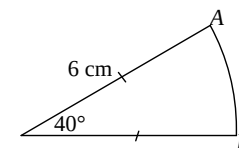
Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) Factorise $x^3 - 8y^3$. 1
- (b) Solve $|2x + 1| = 5$. 2
- (c) Express $\frac{1}{3 - \sqrt{2}}$ with a rational denominator. 2
- (d) Differentiate $3e^{x^2+1}$. 2
- (e) Differentiate $\frac{x^2}{5x+1}$. 2
- (f) Find a primitive of $\cos 2x$. 1
- (g) Find the exact value of $\int_0^1 \frac{x}{x^2+1} dx$. 3
- (h) The angle of a sector in a circle of radius 6 cm is 40° , as shown in the diagram below. 2

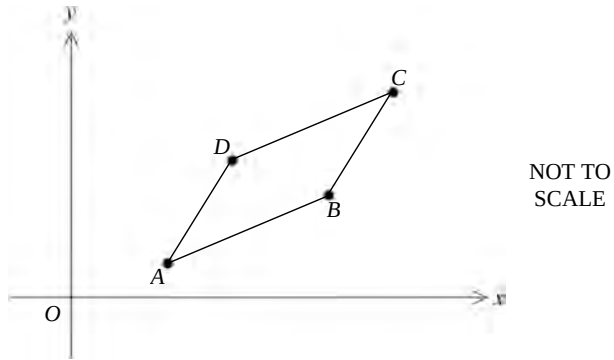


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Find the exact length of arc AB .

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) The points $A(3,1)$, $B(8,3)$, $C(10,6)$ and $D(5,4)$ are the vertices of a parallelogram.



- (i) Find the length of interval BC . 1
- (ii) Show that the equation of BC is $3x - 2y - 18 = 0$. 2
- (iii) Find the perpendicular distance from D to BC . 2
- (iv) Hence, or otherwise, find the area of parallelogram $ABCD$. 1
- (b) 50 tickets are sold in a raffle. The raffle has two prizes, with the winning ticket not being replaced after each draw. Don has bought 4 tickets in the raffle.
- (i) Find the probability Don wins only one prize. 2
- (ii) Find the probability Don does not win a prize. 1
- (iii) Find the probability Don wins at least one prize. 2
- (c) A parabola has equation $x^2 - 6x + 8y + 17 = 0$.
- (i) Find the coordinates of the vertex of the parabola. 2
- (ii) Sketch the parabola, clearly showing the vertex, focus and directrix. 2

Question 13 (15 marks) Use a SEPARATE writing booklet.

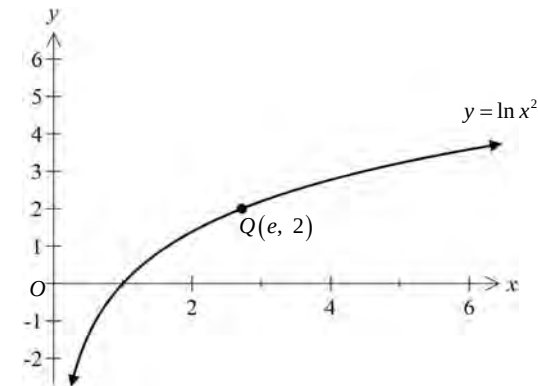
- (a) The first three terms of an arithmetic series are -2 , 0.5 and 3 .

- (i) Find the 24th term. 2
- (ii) Find the sum of the first 24 terms. 1

- (b) Consider the curve $y = x^3 + 3x^2 - 9x - 2$.

- (i) Find any stationary points and determine their nature. 4
- (ii) Find the coordinates of any point(s) of inflexion. 2
- (iii) Sketch the curve labelling the stationary points, point of inflexion and y-intercept. 2

- (c) The point $Q(e, 2)$ lies on the curve $y = \ln x^2$.



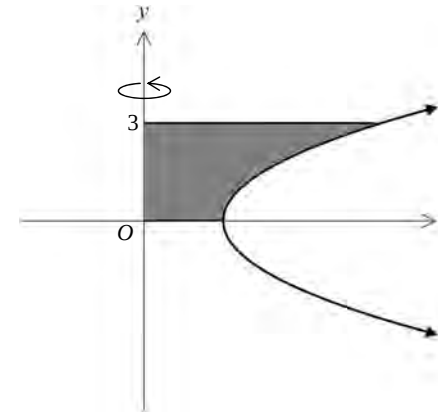
- (i) Use Simpson's rule with 5 function values to approximate $\int_1^5 \ln x^2 dx$, correct to two decimal places. 2
- (ii) Find the equation of the tangent to the curve $y = \ln x^2$ at point Q . 2

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) Find the values of k for which the equation $y = 3x^2 + kx - 4k$ is positive definite. 2
- (b) A particle moves in a straight line. It's velocity, V m/s, is given by the function $V = -3t^2 + 4t + 4$. Initially the particle is 3 metres to the right of the origin.
- (i) Find when the particle is at rest. 2
- (ii) Find an expression for the displacement of the particle after t seconds. 2
- (iii) Find the displacement of the particle after 4 seconds. 1
- (iv) Find the total distance travelled by the particle in the first 4 seconds. 2
- (c) Henry borrows \$820 000 to purchase an apartment. The loan is to be repaid at a reducible interest rate of 4.8% p.a. The loan is repaid in monthly instalments of \$ M .
- (i) Show that the amount owing after 2 months, A_2 , is given by 1
- $$A_2 = 820\,000(1.004)^2 - M(1 + 1.004).$$
- (ii) If the length of the loan is 25 years, show that the value of M , the monthly repayment, is \$4698.58. 2
- (iii) Instead, Henry makes monthly repayments of \$5100. 3
After how many months will he have fully repaid the loan?

Question 15 (15 marks) Use a SEPARATE writing booklet.

- (a) A parabola has equation $y^2 = x - 1$. 3



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The area bounded by the curve $y^2 = x - 1$, the y -axis and the lines $y = 0$ and $y = 3$ is rotated about the y -axis to form a solid.

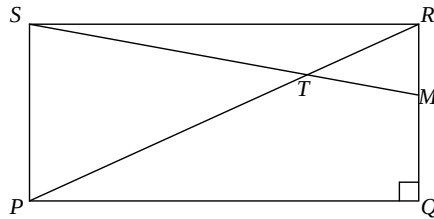
Find the volume of the solid.

- (b) The value, \$ V , of a car after t years is given by the equation $V = Ae^{-kt}$, where A and k are positive constants which depend on the make and model of the car.
- (i) Kate buys a new hatchback for \$15 000. After 1 year her car is valued at \$13 000. 2
- Show that, for Kate's car, $k = 0.143$ correct to 3 significant figures.
- (ii) At the same time Brian buys a new sports car. Its value is given by the equation $V = 20\,000e^{-0.2t}$. 3
- Find how long it is before Brian's and Kate's cars have the same value.
- (iii) At what rate is the value of Brian's car decreasing after 3 years? 2

Question 15 continues on the following page

Question 15 (continued)

- (c) $PQRS$ is a rectangle and lines PR and SM intersect at T .
Point M divides RQ in the ratio $1 : 2$.



NOT TO
SCALE

Copy or trace the diagram into your writing booklet.

- (i) Show that $\triangle MTR \parallel \triangle STP$.

2

- (ii) Given $PR = 30$ cm, find the length of RT .

3

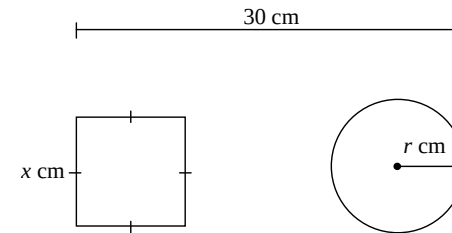
End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

(a) Evaluate $\int_{-\log_e 3}^0 \frac{4}{e^{2x}} dx$.

3

- (b) A 30 cm length of wire is used to make two frames. The wire is to be cut into two parts. One part is bent into a square of side x cm and the remaining length is bent into a circle of radius r cm.



- (i) The circumference of a circle, C , is found using the formula $C = 2\pi r$.
Show that the expression for r in terms of x is $r = \frac{15 - 2x}{\pi}$.

1

- (ii) Show that the combined area, A , of the two shapes can be written as
 $A = \frac{(4 + \pi)x^2 - 60x + 225}{\pi}$.

2

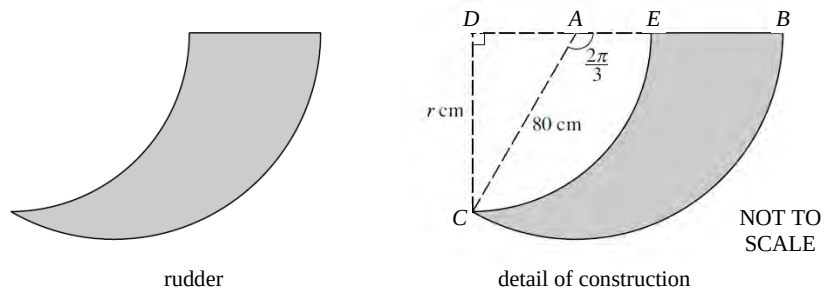
- (iii) Find the value of x for which the combined area of the two frames will be minimised. Give your answer correct to 2 significant figures.

3

Question 16 continues on the following page

Question 16 (continued)

(c) The diagram below shows the cross-section of a rudder.



BC is an arc of a circle with centre A and radius 80 cm. $\angle CAB = \frac{2\pi}{3}$.
 EC is an arc of a circle with centre D and radius r cm. $\angle CDE$ is a right angle.

- (i) Show the area of sector ABC is $\frac{6400\pi}{3}$ cm². 1
- (ii) Show that $r = 40\sqrt{3}$. 2
- (iii) Hence, or otherwise, calculate the area of the cross-section of the rudder, correct to two decimal places. 3

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YEAR 12 TRIAL - 2 UNIT - 2016

MULTIPLE CHOICE

①

- ① C ② A ③ D ④ C ⑤ B
⑥ A ⑦ D ⑧ C ⑨ C ⑩ A

① $x \times 1.1 = 1364$

$x = 1240$

C

② $x^2 + 5x - 2$

$\alpha\beta = -\frac{2}{1} \quad \alpha + \beta = \frac{-5}{1}$
 $= -2 \quad = -5$

$\alpha\beta - (\alpha + \beta) = -2 - (-5)$
 $= 3$

A

③ centre $(2, -3)$ radius 5

$(x-2)^2 + (y+3)^2 = 5^2$

$(x-2)^2 + (y+3)^2 = 25$

D

④ concave up, positive gradient

C

⑤ $2\sin x = -\sqrt{3}$

$\sin x = -\frac{\sqrt{3}}{2}$

$x = \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$
 $= -\frac{\pi}{3}$

$x = \pi + \frac{\pi}{3}, 2\pi - \frac{\pi}{3}$

$= \frac{4\pi}{3}, \frac{5\pi}{3}$

B



⑥ $a = \log_5 2 \quad b = \log_5 3$

$\log_5 36 = \log_5 4 + \log_5 9$
 $= \log_5 2^2 + \log_5 3^2$
 $= 2\log_5 2 + 2\log_5 3$
 $= 2(\log_5 2 + \log_5 3)$
 $= 2(a+b)$

A

⑦ $S_{\infty} = \frac{a}{1-r} \quad r = \frac{3x}{4}$
 $140 = \frac{4}{1-2x} \quad = 2x$

$140 - 280x = 4$

$-280x = -136$

$x = \frac{17}{35}$

D

⑧ $A(3,6) \quad A = \int_3^3 -2x^2 + 8x - 2x$
 $= 0 \int_3^3 -2x^2 + 6x$

C

⑨ $P(\text{even or } 3) = P(\text{even}) + P(x=3) - P(\text{both})$
 $= \frac{10}{20} + \frac{6}{20} - \frac{3}{20}$
 $= \frac{13}{20}$

C

⑩ $y = \sin\left[2\left(x - \frac{\pi}{6}\right)\right] \quad A = 2$
 $= \sin\left(2x - \frac{\pi}{3}\right) \quad B = \frac{\pi}{3}$

A

QUESTION 11

②

a) $x^3 - 8y^3 = x^3 - (2y)^3$
 $= (x-2y)(x^2 + 2xy + 4y^2)$

b) $|2x+1| = 5$

$2x+1 = 5$

$2x = 4$

$x = 2$

or $2x+1 = -5$

$2x = -6$

$x = -3$

c) $\frac{1}{3-\sqrt{2}} \times \frac{3+\sqrt{2}}{3+\sqrt{2}} = \frac{3+\sqrt{2}}{3^2 - (\sqrt{2})^2}$
 $= \frac{3+\sqrt{2}}{9-2}$
 $= \frac{3+\sqrt{2}}{7}$

d) $\frac{d}{dx} 3e^{x^2+1} = 3e^{x^2+1} \times 2x$
 $= 6xe^{x^2+1}$

e) $\frac{d}{dx} \frac{x^2}{5x+1} = \frac{vu' - uv'}{v^2}$
 $= \frac{2x(5x+1) - 5x^2}{(5x+1)^2}$
 $= \frac{10x^2 + 2x - 5x^2}{(5x+1)^2}$
 $= \frac{5x^2 + 2x}{(5x+1)^2}$
 $= \frac{x(5x+2)}{(5x+1)^2}$

$u = x^2$
 $u' = 2x$

$v = 5x+1$
 $v' = 5$

f) $\int \cos 2x = \frac{\sin 2x}{2} + c$

③

$$\begin{aligned}
 \text{g) } \int_0^1 \frac{x}{x^2+1} dx &= \left[\frac{1}{2} \ln|x^2+1| \right]_0^1 \\
 &= \frac{1}{2} \left[\ln|1^2+1| - \ln|0^2+1| \right] \\
 &= \frac{1}{2} [\ln 2 - \ln 1] \\
 &= \frac{1}{2} \ln 2 \\
 &= \ln \sqrt{2}
 \end{aligned}$$

$$\begin{aligned}
 \text{h) } 40^\circ &= 40 \times \frac{\pi}{180} \text{ radians} \\
 &= \frac{2\pi}{9} \\
 \text{Arc length} &= r\theta \\
 &= 6 \times \frac{2\pi}{9} \\
 &= \frac{4\pi}{3} \text{ cm}
 \end{aligned}$$

QUESTION 12

④

$$\begin{aligned}
 \text{a) } B(8,3), C(10,6) \\
 BC &= \sqrt{(10-8)^2 + (6-3)^2} \\
 &= \sqrt{4+9} \\
 &= \sqrt{13}
 \end{aligned}$$

$$\begin{aligned}
 \text{ii) } m_{BC} &= \frac{6-3}{10-8} \\
 &= \frac{3}{2}
 \end{aligned}$$

$$\begin{aligned}
 \therefore y-6 &= \frac{3}{2}(x-10) \\
 2y-12 &= 3(x-10) \\
 2y-12 &= 3x-30 \\
 0 &= 3x-2y-18
 \end{aligned}$$

$$\begin{aligned}
 \text{iii) } d &= \frac{|ax+by+c|}{\sqrt{a^2+b^2}} \quad \text{where } a=3, b=-2, c=-18, D(5,4) \\
 &= \frac{|3 \times 5 + (-2) \times 4 + (-18)|}{\sqrt{3^2 + (-2)^2}} \\
 &= \frac{|15 - 8 - 18|}{\sqrt{13}} \\
 &= \frac{|-11|}{\sqrt{13}} \\
 &= \frac{11}{\sqrt{13}}
 \end{aligned}$$

$$\begin{aligned}
 \text{iv) } A &= bh \\
 &= \sqrt{13} \times \frac{11}{\sqrt{13}} \\
 &= 11 \text{ u}^2
 \end{aligned}$$

⑤

$$b) i) P(w_1) + P(w_2) = \frac{4}{50} \times \frac{46}{49} + \frac{46}{50} \times \frac{4}{49}$$

$$= \frac{184}{1225}$$

$$ii) P(L) = \frac{46}{50} \times \frac{45}{49}$$

$$= \frac{207}{245}$$

$$iii) P(\text{at least one}) = 1 - \frac{207}{245}$$

$$= \frac{38}{245}$$

$$c) x^2 - 6x + 8y + 17 = 0$$

$$i) x^2 - 6x = -8y - 17$$

$$x^2 - 6x + 9 = -8y - 17 + 9$$

$$(x-3)^2 = -8y - 8$$

$$(x-3)^2 = -8(y+1)$$

$$\therefore \text{vertex } (3, -1)$$

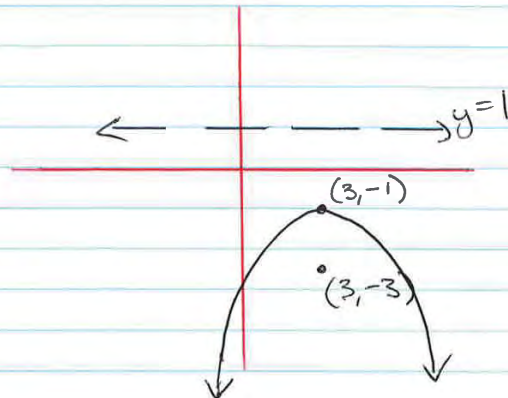
$$ii) \text{vertex } (3, -1)$$

$$a = 2$$

$$\text{focus } (3, -3)$$

$$\text{directrix } y = 1$$

$$\text{same face parabola}$$



Question 13

⑥

$$a) -2, 0.5, 3$$

$$i) a = -2, d = 3 - 0.5 = 2.5$$

$$T_{24} = -2 + (24-1) \times 2.5$$

$$= 55.5$$

$$ii) S_n = \frac{n}{2}(a+L)$$

$$= \frac{24}{2}(-2 + 55.5)$$

$$= 642$$

$$b) y = x^3 + 3x^2 - 9x - 2$$

$$i) y' = 3x^2 + 6x - 9$$

$$y'' = 6x + 6$$

stat. points when $y' = 0$

$$0 = 3x^2 + 6x - 9$$

$$0 = x^2 + 2x - 3$$

$$0 = (x+3)(x-1)$$

$$\therefore x = -3, x = 1$$

$$\text{when } x = -3, y = (-3)^3 + 3(-3)^2 - 9(-3) - 2 = 25$$

$$\text{when } x = 1, y = (1)^3 + 3(1)^2 - 9(1) - 2 = -7$$

$$\text{at } x = -3, y'' = 6(-3) + 6 = -12$$

$$< 0$$

$$\therefore (-3, 25) \text{ is max t.p.}$$

$$\text{at } x = 1, y'' = 6(1) + 6 = 12$$

$$> 0$$

$$\therefore (1, -7) \text{ is min t.p.}$$

(7)

ii) pt of inflexion when $y''=0$

$$0 = 6x + 6$$

$$-6 = 6x$$

$$x = -1$$

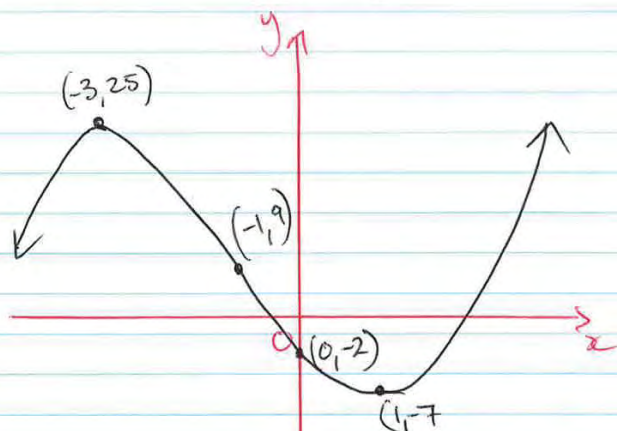
when $x = -1$, $y = 9$

x	-2	-1	0
y''	-6	0	6

$\therefore$ change in concavity

$\therefore$ point of inflexion at $(-1, 9)$

iii)



(8)

e) i)

x	1	2	3	4	5
y	$\ln 1$	$\ln 4$	$\ln 9$	$\ln 16$	$\ln 25$

$$\int_1^5 \ln x^2 dx \doteq \frac{3-1}{6} (\ln 1 + 4 \times \ln 4 + \ln 9) + \frac{5-3}{6} (\ln 9 + 4 \times \ln 16 + \ln 25)$$

$$\doteq 8.08295 \dots$$

$$\doteq 8.08$$

ii) $y = \ln x^2$ at $(e, 2)$

$$y' = \frac{1}{x^2} \times 2x$$

$$= \frac{2}{x}$$

at $x = e$, $y' = \frac{2}{e}$

$$\therefore m = \frac{2}{e} \quad \text{At } (e, 2)$$

$$y - 2 = \frac{2}{e}(x - e)$$

$$ey - 2e = 2(x - e)$$

$$ey - 2e = 2x - 2e$$

$$0 = 2x - ey$$

QUESTION 14

(9)

a) $y = 3x^2 + kx - 4k$

$$\Delta < 0$$

$$k^2 - 4 \times 3 \times -4k < 0$$

$$k^2 + 48k < 0$$

$$k(k + 48) < 0$$

critical points inequality

$$k(k + 48) = 0$$

$$k = 0 \quad k = -48$$



$$\therefore -48 < k < 0$$

b) $V = -3t^2 + 4t + 4$

at rest when $V = 0$

$$0 = -3t^2 + 4t + 4$$

$$0 = -3t^2 + 6t - 2t + 4$$

$$0 = -3t(t-2) - 2(t-2)$$

$$0 = (t-2)(-3t-2)$$

$$\therefore t-2=0 \quad -3t-2=0$$

$$t=2 \quad t=-\frac{2}{3}$$

$\therefore$ stationary when $t=2$.

p-12
3 4
F 6, -2

(10)

ii) displacement $= \int v$.

$$\int -3t^2 + 4t + 4 \, dt = -\frac{3t^3}{3} + \frac{4t^2}{2} + 4t + c$$

$$= -t^3 + 2t^2 + 4t + c$$

when $t=0$, $x=3$

$$\therefore 3 = -0^3 + 2 \cdot 0^2 + 4 \cdot 0 + c$$

$$c = 3$$

$$\therefore x = -t^3 + 2t^2 + 4t + 3$$

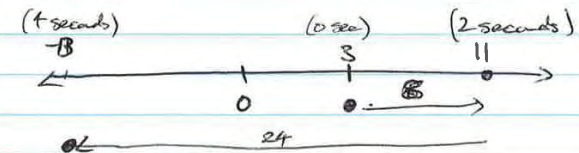
iii) $x = -4^3 + 2 \cdot 4^2 + 4 \cdot 4 + 3$

$$= -13$$

iv) $V=0$ at $t=2$.

$$\therefore \text{at } t=2 \quad x = -2^3 + 2 \cdot 2^2 + 4 \cdot 2 + 3$$

$$= 11$$



$$\therefore \text{total distance} = (11-3) + (11-(-13))$$

$$= 8 + 24$$

$$= 32$$

(11)

$$i) A_1 = 820000 \times (1 + 0.004)^1 - M$$

$$r = \frac{0.048}{12} = 0.004$$

$$= 820000 \times 1.004 - M$$

$$A_2 = A_1 \times 1.004 - M$$

$$= (820000 \times 1.004 - M) \times 1.004 - M$$

$$= 820000 \times 1.004^2 - M \times 1.004 - M$$

$$= 820000 \times 1.004^2 - M(1 + 1.004)$$

$$ii) A_n = 820000 \times 1.004^n - M(1 + 1.004 + \dots + 1.004^{n-1})$$

loan repaid after 25 years, $\therefore n = 300$

$$\therefore A_{300} = 0$$

$$0 = 820000 \times 1.004^{300} - M(1 + 1.004 + \dots + 1.004^{299})$$

$$0 = 820000 \times 1.004^{300} - M \left(\frac{1(1.004^{300} - 1)}{1.004 - 1} \right)$$

$$0 = 820000 \times 1.004^{300} - M \times 578.0448 \dots$$

$$M = \frac{820000 \times 1.004^{300}}{578.0448 \dots}$$

$$= 4698.5753 \dots$$

$$= \$4698.58$$

(12)

$$iii) A_n = 820000 \times 1.004^n - M(1 + 1.004 + \dots + 1.004^{n-1})$$

$$\text{let } M = 5100, A_n = 0$$

$$0 = 820000 \times 1.004^n - 5100(1 + 1.004 + \dots + 1.004^{n-1})$$

$$0 = 820000 \times 1.004^n - 5100 \left(\frac{1(1.004^n - 1)}{1.004 - 1} \right)$$

$$0 = 820000 \times 1.004^n - 1275000(1.004^n - 1)$$

$$0 = 820000 \times 1.004^n - 1275000 \times 1.004^n + 1275000$$

$$-1275000 = 1.004^n(820000 - 1275000)$$

$$1.004^n = \frac{-1275000}{-455000}$$

$$\log_e 1.004^n = \log_e \left(\frac{255}{91} \right)$$

$$n \times \log_e 1.004 = \log_e \left(\frac{255}{91} \right)$$

$$n = \log_e \left(\frac{255}{91} \right) \div \log_e(1.004)$$

$$n = 258.1158 \dots$$

$\therefore$ Repaid after 259 months.

QUESTION 15

(13)

$$\begin{aligned}
 a) \int_0^3 [f(y)]^2 dy \quad & y^2 = x - 1 \\
 & x = y^2 + 1 \\
 & x^2 = (y^2 + 1)^2 \\
 & x^2 = y^4 + 2y^2 + 1 \\
 \therefore V = \pi r^2 \int_0^3 y^4 + 2y^2 + 1 dy \\
 & = \pi \left[\frac{y^5}{5} + \frac{2y^3}{3} + y \right]_0^3 \\
 & = \pi \times \left[\left(\frac{3^5}{5} + \frac{2 \times 3^3}{3} + 3 \right) - \left(\frac{0}{5} + \frac{0}{3} + 0 \right) \right] \\
 & = \pi \times \frac{348}{5} \\
 & = \frac{348\pi}{5} \text{ m}^3
 \end{aligned}$$

$$\begin{aligned}
 b) i) V &= Ae^{-kt} \\
 \text{at } t=0, V &= 15000 \quad \therefore 15000 = Ae^0 \\
 & \quad A = 15000 \\
 \text{at } t=1, V &= 13000 \\
 \therefore 13000 &= 15000 e^{-k} \\
 \frac{13000}{15000} &= e^{-k} \\
 \log_e \left(\frac{13}{15} \right) &= -k \\
 -0.1431 &= -k \\
 k &= 0.143
 \end{aligned}$$

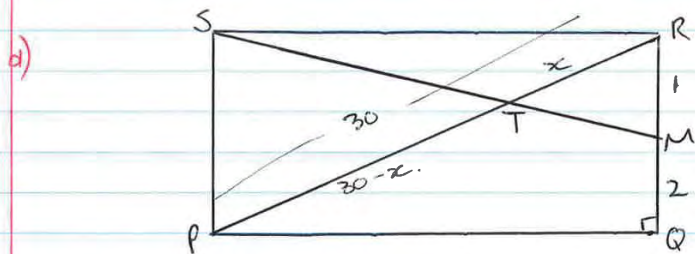
(14)

$$\begin{aligned}
 ii) V &= 20000e^{-0.2t} \quad V = 15000e^{-0.143t} \\
 20000e^{-0.2t} &= 15000e^{-0.143t} \\
 4e^{-0.2t} &= 3e^{-0.143t} \\
 \frac{4e^{-0.2t}}{3e^{-0.143t}} &= 1 \\
 \frac{4e^{-0.2t - (-0.143t)}}{3} &= 1 \\
 \frac{4e^{-0.057t}}{3} &= 1 \\
 e^{-0.057t} &= \frac{3}{4} \\
 \log_e \left(\frac{3}{4} \right) &= -0.057t \\
 t &= \frac{\log_e \left(\frac{3}{4} \right)}{-0.057} \\
 t &= 5.047 \dots
 \end{aligned}$$

$\therefore$ after 5 years

$$\begin{aligned}
 iv) V &= 20000e^{-0.2t} \\
 \frac{dV}{dt} &= 20000e^{-0.2t} \times -0.2 \\
 &= -4000e^{-0.2t} \\
 \text{at } t=3 \quad \frac{dV}{dt} &= -4000e^{-0.2 \times 3} \\
 &= -4000e^{-0.6} \\
 &= -2195.25
 \end{aligned}$$

(15)



- i) In $\triangle MTR$ & $\triangle STP$
- $\angle MTR = \angle STP$ (vertically opposite $\angle$ s are equal)
 - $\angle TMR = \angle TPS$ (alternate $\angle$ s are equal, $SP \parallel RQ$)
 - opposite sides of a rectangle are equal
- $\therefore \triangle MTR \equiv \triangle STP$ (equiangular)

- ii) let $TR = x$, $\therefore TP = 30 - x$

$$\therefore \frac{x}{30-x} = \frac{1}{3} \quad \frac{RT}{PT} = \frac{RM}{PS}$$

$$3x = 30 - x$$

$$4x = 30$$

$$x = 7.5$$

$$\therefore RT = 7.5 \text{ cm}$$

QUESTION 16

(16)

a)

$$\int_{-\log_e 3}^0 \frac{4}{e^{2x}} dx = \int_{-\log_e 3}^0 4e^{-2x} dx$$

$$= 4 \left[\frac{e^{-2x}}{-2} \right]_{-\log_e 3}^0$$

$$= 4x \left[\frac{e^{-2x \cdot 0}}{-2} - \frac{e^{-2x \cdot \log_e 3}}{-2} \right]$$

$$= 4x \left[\frac{1}{-2} + \frac{e^{2 \log_e 3}}{2} \right]$$

$$= 4x \left[-\frac{1}{2} + \frac{9}{2} \right]$$

$$= 4x(4)$$

$$= 16$$

b) i)

$$C = 2\pi r$$

$$C = 30 - 4x$$

$$\therefore 30 - 4x = 2\pi r$$

$$r = \frac{30 - 4x}{2\pi}$$

$$r = \frac{15 - 2x}{\pi}$$

(17)

$$i) A = x^2 + \pi r^2$$

$$= x^2 + \pi x \left(\frac{15-2x}{\pi} \right)^2$$

$$= x^2 + \pi x \left(\frac{15^2 - 60x + 4x^2}{\pi^2} \right)$$

$$= x^2 + \frac{225 - 60x + 4x^2}{\pi}$$

$$= \frac{\pi x^2 + 225 - 60x + 4x^2}{\pi}$$

$$= \frac{(4+\pi)x^2 - 60x + 225}{\pi}$$

$$iii) A = \frac{(4+\pi)}{\pi} x^2 - \frac{60}{\pi} x + 225$$

$$\frac{dA}{dx} = \frac{2(4+\pi)}{\pi} x - \frac{60}{\pi}$$

$$\frac{d^2A}{dx^2} = \frac{2(4+\pi)}{\pi} > 0, \therefore \text{always min. t.p.}$$

$$\text{let } \frac{dA}{dx} = 0 \quad 0 = \frac{2(4+\pi)}{\pi} x - \frac{60}{\pi}$$

$$\frac{60}{\pi} = \frac{2(4+\pi)}{\pi} x$$

$$60 = 2(4+\pi)x$$

$$x = \frac{60}{2(4+\pi)}$$

$$x = \frac{30}{4+\pi}$$

$$\frac{d^2A}{dx^2} > 0$$

$\therefore x = 4.2$ will give minimum area

(18)

$$c) i) A = \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2} \times 80^2 \times \frac{2\pi}{3}$$

$$= \frac{6400\pi}{3}$$

$$ii) \angle CAD = \pi - \frac{2\pi}{3} \quad \boxed{102} \quad \frac{2\pi}{3} = \frac{2\pi}{3} \times \frac{180}{\pi} \text{ degrees}$$

$$= \frac{4\pi}{3} \text{ (straight } \angle \text{ is } 180^\circ) \quad = 120^\circ$$

$$\therefore \sin \frac{\pi}{3} = \frac{r}{80}$$

$$r = 80 \sin \frac{\pi}{3}$$

$$r = 80 \times \frac{\sqrt{3}}{2}$$

$$= 40\sqrt{3}$$

$$\angle CAD = 180 - 120$$

$$= 60^\circ \text{ (straight } \angle \text{ is } 180^\circ)$$

$$\sin 60 = \frac{r}{80}$$

$$r = 80 \sin 60$$

$$r = 40\sqrt{3}$$

$$iii) \text{Area of } \triangle ADC \quad \boxed{102}$$

$$AD = \sqrt{80^2 - (40\sqrt{3})^2}$$

$$= \sqrt{1600}$$

$$= 40$$

$$\angle ACD = 180 - 90 - 60$$

$$= 30^\circ$$

$$\therefore \text{Area } \triangle ADC = \frac{1}{2} \times 40 \times 40\sqrt{3}$$

$$= 800\sqrt{3}$$

$$\therefore \text{Area } \triangle ADC = \frac{1}{2} \times 80 \times 40\sqrt{3} \times \sin 30$$

$$= 1600\sqrt{3} \times \frac{1}{2}$$

$$= 800\sqrt{3}$$

$$\therefore \text{Total Area} = \text{Area Sector ABC} + \text{Area } \triangle ADC - \text{Area Quadrant DEC}$$

$$= \frac{6400\pi}{3} + 800\sqrt{3} - \frac{1}{4} \times \pi \times (40\sqrt{3})^2$$

$$= \frac{6400\pi}{3} + 800\sqrt{3} - 1200\pi$$

$$= 4317.7937$$

$$= 4317.79 \text{ cm}^2$$